

- 15** *Use of the Data Booklet is relevant to this question.*

An electrolysis cell is set up with an anode made of an alloy of 75% Cu and 25% Ni by mass and a cathode made of pure Cu. The electrolyte is a solution which contains 1.0 mol dm^{-3} of $\text{NiSO}_4(\text{aq})$ and 1.0 mol dm^{-3} of $\text{CuSO}_4(\text{aq})$.

A current of 0.50 A is passed through the cell for 1.5 hours . The cathode increased in mass by 0.88 g .

What value of Avogadro's constant does **these figures** give?

A 6.02×10^{23}

B 6.09×10^{23}

C 8.12×10^{23}

D 1.22×10^{24}